

NGAME Churn Persistence Index (CPI) and Dollar-Weighted CPI (wCPI): Full Derivation

Per-transaction daily measures rolled up into composite CPI / wCPI and scored against a trained baseline

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NGAME CPI / wCPI — Full Derivation

This document gives the end-to-end computation for each transaction type's daily measures — the μ baseline and the persistence index that replaced the misused φ coefficient — rolled up into a composite CPI and a dollar-weighted wCPI, then scored against the trained distribution. NGAME tracks 17 transaction types plus Vendors ($\varphi_1 \dots \varphi_{18}$).

Why the original φ was replaced

The legacy metric used the Matthews / φ correlation formula with the contingency cell d hard-coded to 0:

$$\varphi = \frac{ad - bc}{\sqrt{(a+b)(c+d)(a+c)(b+d)}}.$$

With $d = 0$ the numerator reduces to $-bc$, so the output is confined to $[-1, 0]$ — counterintuitive, not auditor-explainable, and impossible to threshold without training variance:

$$\varphi = \frac{-bc}{\sqrt{(a+b)(c)(a+c)(b)}} \leq 0.$$

The revision below restores a proper $[0, 1]$ persistence measure.

1. Daily set-up — the per-type contingency counts

For each transaction type k , compare yesterday's URI set Y_k against today's URI set T_k .

Symbol	Definition	Meaning
a_k	$ Y_k \cap T_k $	Present in both snapshots (retained)
b_k	$ Y_k \setminus T_k $	Yesterday only (removed)
c_k	$ T_k \setminus Y_k $	Today only (new additions)
n_k	$a_k + b_k + c_k$	Total unique transactions across both days

2. Daily per-type measures (the revised $\varphi \rightarrow \text{CPI}$)

The Churn Persistence Index is the **Jaccard similarity** of the two populations, with two directional companions that conserve to 1:

$$\boxed{\text{CPI}_k = \frac{a_k}{a_k + b_k + c_k} = \frac{a_k}{n_k}}$$

$$\text{AdditionRate}_k = \frac{c_k}{n_k}, \quad \text{DeletionRate}_k = \frac{b_k}{n_k},$$

$$\text{CPI}_k + \text{AdditionRate}_k + \text{DeletionRate}_k = 1.0.$$

If $n_k = 0$ (no transactions of type k in either snapshot), then $\text{CPI}_k = 1.0$ by convention (“absence is stable”).

The triplet $(\text{CPI}_k, \text{AdditionRate}_k, \text{DeletionRate}_k)$ characterises the day’s dynamics for type k with no information loss. A **low** CPI (high churn) is the primary fraud concern; a **high** AdditionRate flags bulk insertion; a **high** DeletionRate flags retroactive deletion.

Worked example — single-day CPI

Customers (φ_1): one vendor added.

$$\text{Before: } a = 50, b = 0, c = 0 \Rightarrow \text{CPI} = \frac{50}{50} = 1.000$$

$$\text{After: } a = 50, b = 0, c = 1 \Rightarrow \text{CPI} = \frac{50}{51} = 0.980$$

One added URI is a small move; a bulk insert (e.g. 10 vendors in a day) drives CPI down sharply and lifts the AdditionRate z -score.

Partial churn (illustrative), $a = 2, b = 2, c = 1, n = 5$:

$$\text{CPI} = \frac{2}{5} = 0.400, \quad \text{DeletionRate} = \frac{2}{5} = 0.400, \quad \text{AdditionRate} = \frac{1}{5} = 0.200,$$

with $\text{CPI} + \text{AdditionRate} + \text{DeletionRate} = 1.000$, confirming the conservation identity.

3. Dollar-weighted CPI (wCPI)

Unweighted CPI treats a \$50 expense identically to a \$500{,}000 wire. The weighted form keeps the Jaccard structure but replaces set cardinality with the sum of transaction amounts w_i (taken from `qb:hasTotalAmount`; entity-type records with no dollar amount default to $w_i = \$1$):

$$\text{wCPI}_k = \frac{\sum_{i \in Y_k \cap T_k} w_i}{\sum_{i \in Y_k \cup T_k} w_i} = \frac{\text{retained dollars}}{\text{total dollars in play}}.$$

Worked example — the \$50,000 expense hidden in a normal day (test W2)

Twenty transactions: 19 of \$50 retained, plus one new \$50{,}000 expense, so $a = 19$, $b = 0$, $c = 1$.

$$\text{Unweighted: } \text{CPI} = \frac{19}{20} = 0.9500 \quad (\text{looks normal})$$

$$\text{Weighted: } \text{retained } \$ = 19 \times \$50 = \$950, \quad \text{union } \$ = \$950 + \$50,000 = \$50,950,$$

$$\text{wCPI} = \frac{950}{50,950} = 0.0186 \quad (\text{alarm fires}),$$

$$\text{Divergence} = |\text{CPI} - \text{wCPI}| = |0.9500 - 0.0186| = 0.931 \rightarrow \mathbf{HIGH}.$$

Metric	Value	Reading
Unweighted CPI	0.9500	Looks completely normal (19 of 20 URIs retained)
Dollar-weighted wCPI	0.0186	Exposure is overwhelmingly new dollars
Divergence	0.931	HIGH

Dollar-alarm thresholds on divergence: $> 0.30 = \text{HIGH}$, $> 0.15 = \text{MEDIUM}$. This is the count-blind exposure the unweighted index cannot see.

4. The trained baseline — daily μ and σ per type

Over an N -day training window, each type accumulates a distribution of daily CPI values. The training matrix stores the mean and standard deviation. (The original matrix stored only the mean — the “Average Phi array”; σ had to be added.)

$$\mu_k = \frac{1}{N} \sum_{t=1}^N \text{CPI}_k^{(t)}, \quad \sigma_k = \sqrt{\frac{1}{N-1} \sum_{t=1}^N \left(\text{CPI}_k^{(t)} - \mu_k \right)^2}.$$

AdditionRate and DeletionRate carry their own baselines $(\mu_k^{\text{add}}, \sigma_k^{\text{add}})$ and $(\mu_k^{\text{del}}, \sigma_k^{\text{del}})$. Optional day-of-week stratification computes separate (μ_k, σ_k) per weekday (requires roughly 60–90 training days for a stable σ).

5. Daily anomaly score — z against the trained distribution

$$z_k = \frac{\text{CPI}_k^{\text{today}} - \mu_k}{\sigma_k}.$$

Condition	Level	Statistical meaning
$ z_k \leq 2.0$	NORMAL	Within 95% of training distribution
$2.0 < z_k \leq 3.0$	MEDIUM	Outside 95%, within 99.7%

Condition	Level	Statistical meaning
$ z_k > 3.0$	HIGH	Beyond 3σ — rare under normal operations

Worked example — ChartOfAccounts z -score

With today's CPI = 0.065 against a trained $\mu = 0.049$, $\sigma = 0.004$:

$$z = \frac{0.065 - 0.049}{0.004} = 4.0 \rightarrow \mathbf{HIGH}.$$

A tight σ (0.004) makes even a small absolute move land 4σ out. The same calculation drives the live dashboard's TOP ANOMALIES list (z , deviation level, percent deviation).

6. Composite roll-up — the multi-day sequence (SCPI / weighted SCPI)

A “slow-burn” attack spreads small daily moves that never trip a single-day alarm. The sequence index is the **geometric mean** of the last L daily CPIs (computed in log-space for numerical stability), scored with a $\sqrt{L}$ -tightened z . The dollar path runs identically over the daily wCPI buffer.

$$\text{SCPI}_k(t, L) = \left(\prod_{i=1}^L \text{CPI}_k(t - i + 1) \right)^{1/L},$$

$$z_k^{\text{seq}} = \frac{\text{SCPI}_k - \mu_k}{\sigma_k / \sqrt{L}},$$

using the same μ_k, σ_k from the training matrix — no schema change. The weighted SCPI is the identical construction computed over the daily wCPI buffer.

Slow-burn detection — the payoff. A vendor type drifting at only -2σ /day (MEDIUM at most on any single day) scores $z^{\text{seq}} \approx -4.47$ on day 5 — a HIGH alert the per-day detector would never raise.

Implementation: a rolling buffer persists 18-element daily CPI (and wCPI) records; `ngame_sequence_cpi_agent.py` returns `ready=False` until L days exist, so the existing single-day pipeline is unaffected until day L .

7. The core code that produces the daily triplet

From `_calculate_phi_for_transaction_type` (renamed to CPI) in `ngame_cpi_analysis_agent.py`:

```

a = len(yesterday_uris & today_uris)    # retained
b = len(yesterday_uris - today_uris)    # removed
c = len(today_uris - yesterday_uris)    # added
n = a + b + c                           # total unique

if n == 0:                               # stable by convention
    return {'cpi': 1.0, 'addition_rate': 0.0, 'deletion_rate': 0.0,
            'retained': 0, 'added': 0, 'removed': 0}

return {
    'cpi':          a / n,                # Jaccard persistence score

```

```

    'addition_rate': c / n,          # fraction new
    'deletion_rate': b / n,         # fraction removed
    'retained': a, 'added': c, 'removed': b,
}

```

Sources: chat 5f6dc7aa ($PHI \rightarrow CPI / wCPI$ derivation), chat 2eb9f286 (Training Matrix μ/σ z-score), chat 9f2b1e90 (sequence SCPI composite). The W2 wCPI figures are the agent's verified test output ($\$950$ retained $\div$ $\$50,950$ union = 0.0186).